

Ritesh Kumar

Jamshedpur, India | +91-6206269895 | riteshkumar90359@gmail.com
linkedin.com/in/riteshkumar-tech | github.com/RiteshKumar2e | riteshkr.info | ORCID

Professional Summary

Computer Science graduate with internship experience in software application development, full-stack engineering, and REST API development at TechMantra Global and NIT Jamshedpur. Skilled in building scalable web applications, engineering back-end services, implementing testing workflows, and collaborating using Git-based development practices.

Education

Arka Jain University, Jamshedpur

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.47

Aug. 2022 – May 2026

Experience

TechMantra Global

Web Development Intern

May 2024 – July 2024

Noida (Remote)

- **Software Application Development** — Engineered backend features for a full-stack web application using Node.js, Express.js, MySQL, and REST APIs, implementing 10+ production-ready endpoints with input validation, exception handling, reusable code, and software engineering best practices.
- **Testing, Performance & DevOps** — Reduced API response time by **35%** through SQL query optimization and database indexing, implemented unit and integration tests using Jest, and automated linting, testing, and deployment with GitHub Actions CI/CD.

Machine Vision and Intelligence Lab, NIT Jamshedpur

Research Intern

May 2025 – Present

Jamshedpur (On-Site)

- **Steel Surface Defect Detection**
 - **Machine Learning & Software Development** — Developed a modular Python-based industrial defect detection system using TensorFlow, PyTorch, OpenCV, MobileNetV2, Feature Pyramid Network (FPN), and attention, implementing reusable preprocessing, training, evaluation, and inference pipelines for the 6-class NEU-DET dataset.
 - **Model Design & Optimization** — Designed and optimized lightweight deep learning architectures for steel surface defect classification, achieving high accuracy and **0.85 mAP** while enabling efficient edge deployment through multi-scale feature fusion and attention-based learning.
 - **Stack:** Python, TensorFlow, PyTorch, OpenCV, MobileNetV2, FPN, Deep Learning, Computer Vision, Git.

Projects

QuickFix — AI-Powered Customer Complaint Management System

Live | GitHub

- Built a full-stack complaint resolution platform with an async pipeline of specialized AI services (Orchestrator, Classifier, Sentiment Analyzer, Priority Detector, RAG Engine); sustains 15,900+ daily requests at <200ms response time via a 3-tier LLM fallback (Groq LLaMA 3.3 → Gemini 2.0 Flash → local TF-IDF).
- Implemented Google OAuth 2.0 + email OTP authentication, JWT-based role access control, and Brevo SMTP notifications.
- **Stack:** FastAPI, React 19, Turso (LibSQL), Groq API, Gemini 2.0, JWT, REST APIs.

Community AI Platform

Live | GitHub

- Built a multilingual platform providing access to 50+ government schemes, learning resources, and job opportunities across India, with a multi-model AI pipeline (Groq → Gemini fallback) for source-attributed Q&A and content ingestion via News and YouTube APIs; supports English, Hindi, Bengali, Telugu, and Marathi.
- **Stack:** FastAPI, React, Turso (LibSQL), Groq API, Gemini, Brevo API, JWT.

SOEIT Student Achievement Portal

Live | GitHub

- Designed a full-stack academic platform with three roles (Student / Faculty / Admin) covering achievement verification, certificate uploads, gamified leaderboards, bulk certificate export, and one-click ATS-compliant resume generation.
- **Stack:** Node.js, Express, React 19, Turso (LibSQL), Docker, JWT, Google OAuth, Brevo API.

Technical Skills

Languages: Python, C/C++, JavaScript, SQL, HTML/CSS

Backend: FastAPI, Node.js, Express, SQLAlchemy, REST API design | **Frontend:** React.js, MUI

Databases: PostgreSQL, MySQL, MongoDB, Turso (LibSQL)

AI / ML: TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas, MobileNetV2, FPN, Attention Mechanisms, RAG

Google Technologies: Gemini API, Google OAuth 2.0, Google Colab / Jupyter

Tools & Deployment: Git, GitHub Actions (CI/CD), Docker, JWT, Render, Vercel, VS Code

Coursework: Data Structures & Algorithms, OOP, DBMS

Publications

- **Cross-Enhanced and Attention-Modulated Multi-Scale Feature Fusion for Efficient Industrial Surface Defect Identification**

Ritesh Kumar, K. K. Singh — *Discover Artificial Intelligence*, Springer Nature, 2026. (Under Review)

Achievements

- **IBM Hackathon 2025** — National Finalist; qualified via IBM HackerRank challenge and competed in a 24-hour national round.
- **AutonomousHacks '26** — Offline finalist, selected from 2,000+ participants in the Agentic AI track organized by GDG Gandhinagar.
- **CodeFest, Arka Jain University** — Secured 2nd place in the Engineering Day coding competition.
- **NPTTEL (Database Management Systems)** — Certification completed.

Leadership & Extracurricular

- **President, Code & Compute Society (CCS), Arka Jain University** — Led the university's coding community by organizing technical events, mentoring students, collaborating with faculty, and promoting software development initiatives.
- **Community Lead, Google Developer Groups (GDG) on Campus, AJU** — Organized developer workshops, technical sessions, and hands-on events on Google technologies, fostering collaboration and peer learning.
- **Organizer, Hack Horizon 2.0 (2026)** — Coordinated a university-level hackathon, managing event planning, technical operations, participant engagement, and cross-functional team collaboration.